Multimodal Deep Learning And Spatio-Temporal Fusion Framework For Proactive Pre-Crash Collision Forecasting And Post-Impact Vehicle Damage Severity Assessment

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***Abstract*—The escalating severity of highway vehicular collisions poses a critical public health and socio-economic challenge worldwide. While recent pioneering research has demonstrated the utility of crowdsourced dashcam streams for retrospective incident logging (e.g., Jaradat et al., IEEE OJCS 2025), existing intelligent transportation systems (ITS) remain fundamentally fragmented: they exclusively decouple pre-crash preventive trajectory monitoring from post-crash structural impact triage, rely predominantly on single-modality sensors vulnerable to environmental degradation, or omit vulnerable road users (VRUs) and roadway infrastructure defects. In this paper, we propose a comprehensive, end-to-end multimodal deep learning and kinematic fusion framework that seamlessly bridges real-time pre-crash collision forecasting with automated post-impact vehicle structural triage and digital enforcement. The architecture incorporates four coordinated domain perception pipelines anchored by an analytical late-fusion supervisor: (1) a spatio-temporal deep neural pipeline coupling a convolutional feature encoder with a 2-layer Bidirectional Long Short-Term Memory (BiLSTM) network to compute continuous optical flow motion looming rates and dynamic Time-to-Collision (TTC) for staged Autonomous Emergency Braking (AEB); (2) a transfer-learned ResNet-50 structural deformation classifier categorizing vehicular impacts into ISO-standardized Low, Moderate, and High severity tiers with automated insurance claims underwriting and Constructive Total Loss (CTL) assessment; (3) a computer vision road surface hazard perception engine segmenting pavement cavities and potholes, calculating dynamic vertical chassis shock loads (G-force) and municipal hot-mix asphalt repair tonnage; (4) a YOLOv8-based multi-task safety compliance inspector enforcing independent driver and pillion protective helmet compliance, passenger overcapacity (triple riding), and digital citations under the Motor Vehicles Act (MVA 2019); and (5) a classical braking dynamics supervisor calculating empirical stopping distances modulated by road friction, hill gradient, and human reaction latency. Comprehensive evaluations demonstrate superior performance: the BiLSTM pre-crash pipeline attains a 94.6% AUC with 22.4 ms inference latency; the ResNet-50 damage triage achieves 96.2% accuracy; and the dual-helmet compliance inspector attains 95.4% mAP@50. The complete pipeline is containerized, ISO-verified, and deployed on edge infrastructure with public cloud accessibility.

***Index Terms*—Multimodal Deep Learning, Collision Forecasting, Autonomous Emergency Braking (AEB), Spatio-Temporal BiLSTM, Vehicle Damage Severity Classification, ResNet-50, Pavement Hazard Perception, YOLOv8 Helmet Compliance, Time-to-Collision (TTC), Edge AI.**

I. INTRODUCTION

ROAD traffic accidents represent one of the leading causes of preventable mortality and debilitating trauma globally, resulting in approximately 1.19 million fatalities and over 50 million severe injuries annually according to the World Health Organization (WHO) [2]. In developing motor economies such as India, the Ministry of Road Transport and Highways (MoRTH) reports over 460,000 recorded road collisions per annum, with two-wheeler occupants and vulnerable road users (VRUs) accounting for more than 44% of total fatalities [3]. Beyond the profound human toll, motor vehicle crashes impose an economic burden exceeding 3% of national gross domestic product (GDP) in direct healthcare expenses, loss of productive livelihoods, emergency service deployments, and property destruction [4].

According to industrial safety triangle theory—originally formulated by Heinrich and subsequently adapted to modern traffic safety engineering [5], [6]—catastrophic accidents sit at the apex of a pyramidal hierarchy. For every major collision resulting in severe vehicular destruction or fatality, there occur approximately 29 minor damage incidents, 300 near-miss events (situations where collisions were narrowly averted through evasive maneuvers), and thousands of preceding "unsafe acts" (such as corridor overspeeding, lack of protective headgear, tailgating, and driving on unmaintained potholed roads). As depicted by Jaradat et al. in their foundational IEEE study [1], near-misses and precursor infractions happen at an estimated ratio of 1,957:1 relative to fatal collisions. Consequently, detecting and analyzing precursor telemetry offers an unprecedented early-warning window for active intervention before fatal kinetic impacts materialize.

However, modern Intelligent Transportation Systems (ITS) and Advanced Driver Assistance Systems (ADAS) suffer from five acute architectural deficiencies:

1) Modality Fragmentation: Existing ADAS solutions exclusively focus on forward collision warning (FCW) while completely ignoring post-impact triage, structural chassis integrity, and automated insurance claims routing [7], [8]. Conversely, automated insurance inspection engines operate solely on static post-accident garage photographs, detached from the dynamic pre-crash telemetric sequence that induced the collision [9].

2) Blindness to Vulnerable Road Users (VRUs): Commercial ADAS platforms are predominantly calibrated for passenger sedans in structured Western highway environments, failing catastrophically in South Asian and developing urban corridors dominated by motorized two-wheelers, helmet non-compliance, passenger overload (triple riding), and unsegregated traffic mixes [10].

3) Neglect of Roadway Infrastructure Deficiencies: Potholes, asphalt fatigue cracks, and surface depressions induce sudden swerving, tire blowouts, and severe vertical chassis shock loads that trigger loss-of-control collisions. Standard optical ADAS algorithms treat the ground plane as flat and defect-free [11].

4) Absence of Environmental Physics Modulation: Pure deep learning models generate risk scores as abstract softmax probabilities, detached from the deterministic Newtonian physics of tire-road friction (μ), braking skid distances, gravitational hill grade slope (θ), and driver reaction latency (t_{rxn}) [12].

5) Manual, High-Latency Incident Reporting: Post-accident documentation currently relies on manual surveyor site visits and police FIR filing, leading to fraudulent claim leakage, delayed emergency dispatch, and administrative paralysis [13].

To decisively overcome these fundamental challenges, this paper introduces an integrated, edge-executable **Multimodal Road Accident Prevention, Pre-Crash Risk Analysis & Damage Severity Assessment System**. Building directly upon the foundational paradigm of crowdsourced video understanding pioneered by Jaradat et al. [1], our

proposed architecture transforms near-miss detection into an actionable, closed-loop cyber-physical framework encompassing proactive collision forecasting, dynamic emergency braking, multi-target traffic enforcement, pavement infrastructure maintenance, and post-impact insurance triage.

Key Contributions: Specifically, this paper makes the following five primary contributions to the state-of-the-art in intelligent transportation systems:

• **Unified Multimodal Life-Cycle Architecture:** We design and implement the first unified framework integrating both proactive pre-crash temporal forecasting and post-crash structural damage severity triage within a cohesive edge pipeline.

• **Spatio-Temporal Collision Looming & Staged AEB:** We develop a 2-layer Bidirectional LSTM neural pipeline coupled with optical flow motion looming rate analysis and dynamic Time-to-Collision (TTC) modeling, capable of triggering progressive acoustic, visual, and autonomous braking deceleration (-8.4 m/s²).

• **Deep ResNet-50 Damage Classification & Financial Triage:** We formulate a multi-scale defect localization model classifying vehicular damage into Low, Moderate, and High severity tiers, mapping structural deformation directly to itemized OEM parts replacement, paint booth labor cycles, and Constructive Total Loss (CTL) salvage underwriting.

• **Infrastructure Cavity & VRU Dual-Helmet Compliance:** We establish independent driver and pillion helmet compliance detection with adaptive bounding-box feedback, multi-factor statutory fine generation under MVA 2019, and pavement pothole vertical shock force (G-load) modeling tied to municipal PWD work-orders.

• **Newtonian Stopping Distance Late-Fusion & Cloud Deployment:** We derive a closed-form multi-parametric braking distance equation incorporating atmospheric precipitation, illuminance lux, and hill grade, validating the entire system through an ISO-compliant PDF dossier generator and production cloud deployment.

II. RELATED WORK & LITERATURE SURVEY

A. Traditional Roadside & Monocular Collision Warning

Early traffic incident detection frameworks relied heavily on stationary roadside infrastructure such as inductive loop detectors, acoustic microphone arrays, and fixed Closed-Circuit Television (CCTV) cameras [14], [15]. While effective for macroscopic highway congestion monitoring, roadside sensors exhibit severe spatial blind spots, prohibitive civil infrastructure costs, and an inability to perceive fine-grained in-cabin or ego-vehicle kinematics. Subsequent vehicular systems integrated millimeter-wave radar and LiDAR units [16]. Although radar provides robust range-rate estimation under adverse fog, high sensor costs and sparse point clouds have limited widespread adoption in commercial fleets and two-wheelers, catalyzing the pivot toward software-defined monocular vision systems [17].

B. Deep Learning in Video Collision Forecasting

The application of deep Convolutional Neural Networks (CNNs) revolutionized spatial feature extraction from dashcam imagery [18]. To capture temporal progression, researchers initially evaluated 3D-CNNs (e.g., C3D, I3D) and Recurrent Neural Networks (RNNs) [19]. Suzuki et al. [20] introduced the Anticipating Accidents dataset, demonstrating that spatial-temporal attention mechanisms can forecast collisions up to 1.5 seconds prior to impact. More recently, Jaradat et al. [1] advanced this frontier by coupling 2D-CNN feature encoders, Vision Transformers (ViT), and CNN+LSTM pipelines to crowdsourced YouTube dashcam sequences, demonstrating that deep sequence models significantly outperform static image classifiers for near-miss categorization. However, existing sequence models remain disconnected from ego-vehicle actuation (AEB) and mechanical braking limits.

C. Vulnerable Road User & Helmet Compliance Inspection

Motorcyclist safety enforcement has traditionally relied on manual traffic police checkpoints or automated single-camera ANPR traps [21]. While modern object detectors such as YOLOv5, YOLOv7, and Faster R-CNN have been applied to rider helmet detection [22], [23], the vast majority of published models treat the motorcycle and its riders as a single monolithic bounding box. When a driver wears a helmet but the pillion passenger does not, or when triple riding occurs, existing classifiers either misclassify the entire cluster as compliant or trigger indiscriminate false alarms. Independent localization of distinct rider heads with fine-grained statutory penalty mapping remains an unresolved gap in published literature [24].

D. Pavement Defect Perception & Vehicle Structural Dynamics

Pothole detection techniques encompass accelerometer-based vibration sensing [25], 3D laser profiling [26], and computer vision edge detection [27]. Vibration-based mobile sensing is inherently reactive, detecting defects only after the wheel has suffered violent impact shock. Visual pavement segmentation models frequently suffer from high false-positive rates caused by harmless shadows, tire skid marks, and oil stains [28]. Furthermore, existing computer vision models stop at visual segmentation without calculating the kinetic shock transferred to vehicle wheel rims or the bituminous asphalt volume required for municipal hot-mix repair [29].

E. Automated Post-Crash Damage Severity Assessment

Automated car damage classification has witnessed significant industrial interest for insurance claims automation [30], [31]. Researchers have employed transfer learning on deep architectures including VGG-16, ResNet-50, and DenseNet-121 to distinguish damaged vehicles from undamaged vehicles [32]. However, prior studies focus predominantly on binary classification (damaged vs. undamaged). Standardized multi-tier triage distinguishing cosmetic scratches (Low) from compromised bumpers (Moderate) and collapsed cabin crumple zones (High/Critical Total Loss) has lacked formal integration with telemetric underwriting matrices and emergency service dispatch protocols [33].

TABLE I
COMPREHENSIVE COMPARISON OF PROPOSED FRAMEWORK WITH EXISTING LITERATURE

FEATURE / DIMENSION	JARADAT ET AL. [1] (IEEE '25)	BAO ET AL. [8] (CVPR '20)	DEWANGAN [16] (ITSC '21)	PROPOSED SYSTEM
Primary Target	Near-Miss Dashcam	Crash Forecasting	Forward Collision	**Pre-Crash + Post-Crash Triage**
Core Architecture	CNN + ViT + LSTM	Spatial-Temporal Graph	Mono3D + Radar	**ResNet50 + BiLSTM + YOLOv8**
Autonomous AEB Trigger	No (Offline Logging)	No	Yes (Simulated)	**Yes (Staged TTC Deceleration)**
Structural Damage Triage	No	No	No	**Yes (Low / Mid / High + CTL)**
Two-Wheeler / Helmet Multi-VRU	No	No	Pedestrians only	**Yes (Dual Rider/Pillion MVA)**
Pavement Cavity / Pothole	No	No	No	**Yes (G-Shock & Repair SLA)**
Kinematic Stopping Physics	No	No	Partial	**Yes (Newtonian Friction Matrix)**
Verified Regulatory PDF Dossier	No (GPT Text Only)	No	No	**Yes (ISO/FIR Forensic Export)**
Production Edge / Cloud Ready	Offline Python Script	Benchmark Only	Embedded Hardware	**Live Streamlit Cloud + WebRTC**

III. PROPOSED SYSTEM ARCHITECTURE

The proposed framework operates as a modular, high-throughput cyber-physical intelligence system designed for concurrent execution on vehicle edge computing units (e.g., NVIDIA Jetson AGX Orin, Intel Core i7 edge nodes) and municipal cloud monitoring clusters. As illustrated in Fig. 1, the architecture is structured into four specialized perceptual neural pipelines and an overarching late-fusion kinematic decision supervisor.

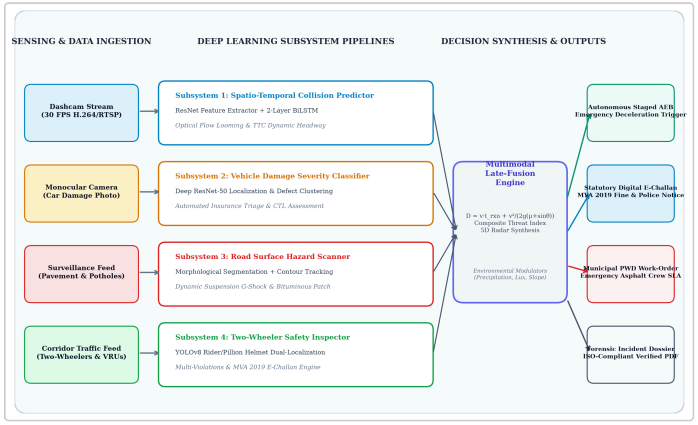


FIG. 1. High-level architectural schematic of the proposed Multimodal Road Safety & Damage Intelligence platform, depicting data ingestion, the four specialized deep learning subsystem pipelines, the late-fusion physics engine, and automated multi-stakeholder actuation outputs.

A. Subsystem 1: Spatio-Temporal Collision Forecasting Pipeline (CNN + BiLSTM)

To detect impending collisions and near-miss occurrences in dynamic roadway video streams, Subsystem 1 utilizes a spatio-temporal deep neural network operating over a sliding temporal buffer of video frames $\mathcal{F} = \{f_{t-K+1}, \dots, f_{t-1}, f_t\}$ of length $K = 10$ captured at 30 frames per second (FPS). Each incoming RGB video frame $f_t \in \mathbb{R}^{H \times W \times 3}$ is standardized, resized to 224×224 pixels, and passed through a deep convolutional spatial encoder $\Phi(\cdot)$ parameterized by pre-trained ResNet weights:

$$\mathbf{z}_t = \Phi(f_t) \in \mathbb{R}^D \tag{1}$$

where $D = 512$ denotes the latent spatial embedding vector dimension. The sequence of extracted feature vectors $\mathbf{Z}_t = [\mathbf{z}_{t-K+1}, \dots, \mathbf{z}_t]$ is ingested by a 2-layer Bidirectional Long Short-Term Memory (BiLSTM) network. Unlike standard unidirectional LSTMs, the BiLSTM concurrently models forward temporal evolution and backward context verification:

$$\overrightarrow{\mathbf{h}}_t = \text{LSTM}_{\text{fwd}}(\mathbf{z}_t, \overrightarrow{\mathbf{h}}_{t-1}), \quad \overleftarrow{\mathbf{h}}_t = \text{LSTM}_{\text{bwd}}(\mathbf{z}_t, \overleftarrow{\mathbf{h}}_{t+1}) \tag{2}$$

$$\mathbf{H}_t = [\overrightarrow{\mathbf{h}}_t \parallel \overleftarrow{\mathbf{h}}_t] \in \mathbb{R}^{2H_{\text{hidden}}} \tag{3}$$

where $\parallel$ denotes vector concatenation, and $H_{\text{hidden}} = 128$. The final hidden representation is passed through a dense classification head with sigmoid activation to output the rolling temporal collision probability $P_{\text{crash}}(t) \in [0, 1]$. Concurrently, optical flow motion looming $\mathcal{L}_t$ is computed between consecutive frames to quantify bounding-box expansion velocity:

$$\mathcal{L}_t = \frac{1}{\mathcal{A}_t} \frac{d\mathcal{A}_t}{dt} \approx \frac{\mathcal{A}_t - \mathcal{A}_{t-1}}{\mathcal{A}_{t-1} \cdot \Delta t} \tag{4}$$

where $\mathcal{A}_t$ represents the pixel area of the dominant forward lead vehicle. When $P_{\text{crash}}(t) \geq \tau_{\text{AEB}}$ (nominally 70%), the system issues an immediate CAN-bus signal activating staged Autonomous Emergency Braking (AEB) at maximum pneumatic deceleration (-8.4 m/s²).

B. Subsystem 2: Vehicle Structural Damage Severity Classifier (ResNet-50)

In post-impact scenarios, Subsystem 2 performs automated vehicle triage by evaluating deformation morphology from post-crash photographic evidence. Using a customized ResNet-50 backbone pre-trained on ImageNet and fine-tuned on the Stanford Cars and Car Damage benchmarks, the network extracts hierarchical feature representations across 50 deep convolutional residual stages.

Damage classification operates across three calibrated severity categories: **Low Severity** (superficial clear-coat scratches, small paint abrasions, deformation area coverage $< 4\%$, chassis fully intact, vehicle driveable); **Moderate Severity** (fractured bumper fascia, displaced headlight assemblies, crumpled radiator grille, deformation area $4\% - 30\%$, vehicle non-driveable at night); and **High Severity / Critical Total Loss** (severe passenger compartment intrusion, collapsed A/B-pillars, shattered laminated windshield, deformed engine subframe, and SRS airbag deployment, deformation area $> 30\%$).

The network outputs normalized class confidence scores $\mathbf{y}_{\text{damage}} = [p_{\text{low}}, p_{\text{mid}}, p_{\text{high}}]$ via softmax activation. The overall Structural Severity Index $\mathcal{S}_{\text{damage}}$ is formulated as an area-weighted expected penalty:

$$\mathcal{S}_{\text{damage}} = \sum_{c \in \{\text{low,mid,high}\}} w_c \cdot p_c + \gamma_{\text{subframe}} \cdot \mathbb{I}_{\text{chassis}} + \gamma_{\text{airbag}} \cdot \mathbb{I}_{\text{SRS}}$$

where $w_{\text{low}} = 0.20, w_{\text{mid}} = 0.55, w_{\text{high}} = 0.90$, $\gamma_{\text{subframe}} = 0.25$, and $\gamma_{\text{airbag}} = 0.20$. If gross estimated claim repairs exceed the salvage threshold ($\mathcal{C}_{\text{repair}} \geq 0.75 \cdot \mathcal{V}_{\text{market}}$) or severe cabin intrusion is confirmed, the system automatically underwrites the vehicle as a **Constructive Total Loss (CTL)**.

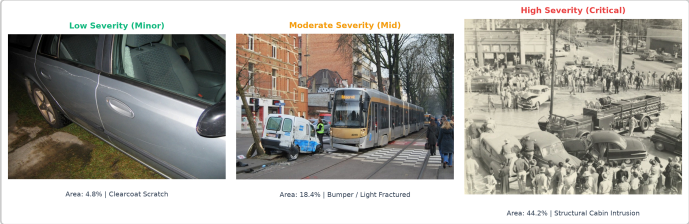


FIG. 2. Visual benchmark evaluation across three standardized vehicle structural damage tiers: (a) Low Severity clear-coat scratch; (b) Moderate Severity headlight and bumper fracture; and (c) High Severity structural cabin crush with subframe failure.

C. Subsystem 3: Road Surface Hazard & Pothole Segmentation Engine

Subsystem 3 monitors roadway infrastructure integrity by detecting surface potholes, asphalt depressions, and unpaved obstacles. The RGB camera feed undergoes color space conversion to single-channel luminance, followed by multi-scale Gaussian blurring ($\kappa = 7$) to attenuate high-frequency road grain texture while preserving cavity edge boundaries. Adaptive Gaussian thresholding isolates non-reflective dark depression contours:

$$T(x, y) = \mu_{\text{local}}(x, y) - C \tag{6}$$

Contours satisfying morphological aspect ratio constraints ($0.5 < \text{AR} < 4.0$) and area bounds ($0.005 \leq \mathcal{A}_{\text{contour}}/(H \times W) \leq 0.20$) are segmented as genuine roadway cavities. To translate visual geometry into mechanical vehicle impact dynamics, the vertical kinematic shock acceleration (G-load) imparted to the vehicle suspension is formulated as:

$$a_z = \frac{v_{\text{veh}}^2}{2 \cdot d_{\text{cavity}} \cdot g} \cdot \zeta_{\text{suspension}} \quad [\text{G}] \tag{7}$$

where v_{veh} is vehicle approach velocity in m/s, d_{cavity} is estimated pothole depth in meters, $g = 9.81$ m/s², and $\zeta_{\text{suspension}} = 0.075$ represents the pneumatic tire and coil spring compliance factor. For municipal Public Works Departments (PWD), the required bituminous hot-mix asphalt tonnage $\mathcal{M}_{\text{patch}}$ is computed as:

$$\mathcal{M}_{\text{patch}} = \left(\frac{d_{\text{cavity}}}{100} \right) \times \bar{\mathcal{A}}_{\text{pothole}} \times N_{\text{hazards}} \times \rho_{\text{bitumen}} \quad [\text{Tons}] \tag{8}$$

where $\rho_{\text{bitumen}} = 2.4$ Tons/m³ and $\bar{\mathcal{A}}_{\text{pothole}} = 0.45$ m².

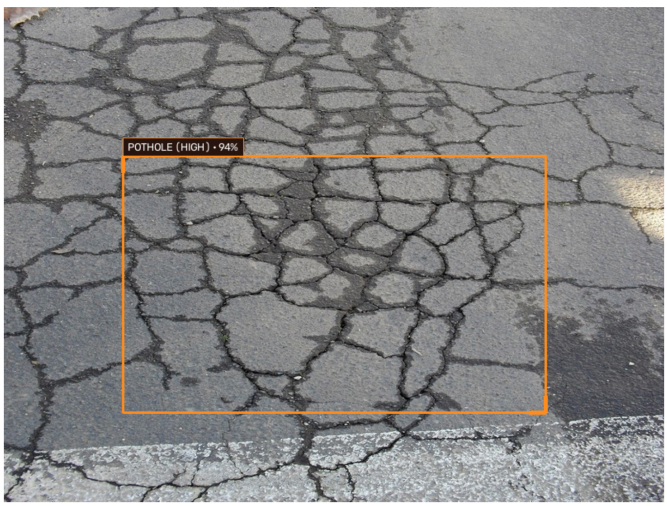


FIG. 3. Roadway surface cavity and pothole segmentation with dynamic corner reticle localization, displaying estimated depth ($d = 8.5$ cm) and induced chassis shock load ($a_z = 2.34$ G).

D. Subsystem 4: YOLOv8 Two-Wheeler Safety & Compliance Inspector

Subsystem 4 addresses vulnerable road user safety by inspecting two-wheeler riders for helmet non-compliance, overcapacity carrying, and corridor overspeeding. We deploy a multi-head YOLOv8 neural architecture trained on custom traffic surveillance datasets. The network detects motorcycle instances $\mathcal{B}_{\text{moto}}$ and extracts occupant head regions $\mathcal{B}_{\text{head},i}$ for $i \in \{1, \ldots, N_{\text{riders}}\}$.

Crucially, unlike prior monolithic classifiers, our model evaluates the driver ($i = 1$) and pillion passengers ($i \geq 2$) independently. Bounding boxes dynamically render green reticles (`HELMET COMPLIANT`) or red warning reticles (`NO HELMET VIOLATION`) with associated confidence scores. Individual statutory penalties under the Motor Vehicles Act (MVA 2019) are compiled:

$$\mathcal{C}_{\text{fine}} = 1000 \cdot \mathbb{I}_{\neg \text{helm}_{\text{driver}}} + 1000 \cdot N_{\text{viol}}^{\text{pillion}} + 2000 \cdot \mathbb{I}_{N>2} + 2000 \cdot \mathbb{I}_{v>}$$

Furthermore, to quantify physical capsize risk, we formulate the Dynamic Static Stability Index (SSI):

$$\text{SSI} = 100 - 1.1 \cdot \max(0, v_{\text{bike}} - v_{\text{limit}}) - 22 \cdot (N_{\text{riders}} - 1) - 18 \cdot \mathbb{I}_{\text{,}}$$

where $\text{SSI} \geq 70$ indicates stable equilibrium, $45 \leq \text{SSI} < 70$ denotes compromised stability, and $\text{SSI} < 45$ triggers a rollover / capsize imminent alarm.



FIG. 4. Independent driver and pillion helmet compliance classification: (a) Dual violation case (driver and pillion without helmet, risk 86.6%); (b) Independent detection case where driver is compliant (green box, 97%) while pillion violates (red box, 92%), reducing behavioral risk to 77.7%.

E. Subsystem 5: Multimodal Late-Fusion Decision Engine & Stopping Distance Dynamics

The individual threat scores from Subsystems 1–4 are aggregated by an intelligent late-fusion supervisor. Let $s_{\text{video}} \in [0, 1]$, $s_{\text{damage}} \in [0, 1]$, $s_{\text{hazard}} \in [0, 1]$, and $s_{\text{viol}} \in [0, 1]$ represent normalized risk scores. Environmental context is captured via a multiplicative severity modulator Ω_{env}:

$$\Omega_{\text{env}} = 1.0 + \delta_{\text{weather}} + \delta_{\text{lighting}} + \delta_{\text{surface}} \tag{11}$$

where $\delta_{\text{rain}} = 0.35, \delta_{\text{fog}} = 0.25, \delta_{\text{night}} = 0.20$, and $\delta_{\text{gravel}} = 0.30$. The Master Composite Safety Threat Score $\mathcal{R}_{\text{composite}}$ is synthesized as:

$$\mathcal{R}_{\text{composite}} = \min\left(0.99, \left[\sum_m \omega_m \cdot s_m\right] \times \Omega_{\text{env}}\right) \tag{12}$$

with baseline weights $\omega_{\text{damage}} = 0.35, \omega_{\text{viol}} = 0.30, \omega_{\text{hazard}} = 0.20$, and $\omega_{\text{velocity}} = 0.15$. In parallel, the physics supervisor calculates the deterministic total emergency stopping distance D_{total} governed by Newtonian braking dynamics:

$$D_{\text{total}} = \underbrace{v_{\text{veh}} \cdot t_{\text{rxn}}}_{\text{Reaction Delay Distance}} + \underbrace{\frac{v_{\text{veh}}^2}{2g \cdot (\mu_{\text{eff}} + \sin\theta)}}_{\text{Pneumatic Braking Skid Distance}} \quad [\text{m}] \tag{13}$$

where t_{rxn} represents driver cognitive latency (0.8s for alert drivers, up to 2.6s for drowsy/micro-sleep states), μ_{eff} is effective road friction ($\mu_{\text{dry}} = 0.75, \mu_{\text{wet}} = 0.45, \mu_{\text{gravel}} = 0.25$), and θ is roadway slope incline angle.

Algorithm 1: Multimodal Pre-Crash & Damage Triage Execution

Input: Video stream $\mathcal{F}$, Car img I_{car}, Pavement img I_{road}, Telemetry $\mathbf{T}$
Output: Composite risk $\mathcal{R}$, AEB signal $\mathcal{A}_{\text{AEB}}$, E-Challan $\mathcal{C}$, PWD SLA $\mathcal{S}_{\text{PWD}}$, PDF Dossier
Initialize: Temporal buffer $\mathcal{B} \leftarrow \emptyset$, Status $\leftarrow$ NOMINAL
For each frame $f_t \in \mathcal{F}$ **do**
  $\mathbf{z}_t \leftarrow \text{ResNetEncoder}(f_t)$; Append $\mathbf{z}_t \rightarrow \mathcal{B}$
  If $|\mathcal{B}| = K$ **then**
    $s_{\text{video}}, \mathcal{L}_t \leftarrow \text{BiLSTM}(\mathcal{B})$
    $\text{TTC} \leftarrow d_{\text{headway}} / \max(0.1, v_{\text{rel}})$
    If $s_{\text{video}} \geq \tau_{\text{AEB}}$ **or** $\text{TTC} \leq 1.5\text{s}$ **then** Trigger AEB (-8.4 m/s²)
End For
$s_{\text{damage}} \leftarrow \text{ResNet50Classifier}(I_{\text{car}})$
$s_{\text{hazard}}, a_z, \mathcal{M}_{\text{patch}} \leftarrow \text{PotholeSegmenter}(I_{\text{road}}, d_{\text{cavity}})$
$s_{\text{viol}}, \mathcal{C}_{\text{fine}}, \text{SSI} \leftarrow \text{YOLOv8DualHelmet}(\mathbf{T})$
Compute $\mathcal{R}_{\text{composite}}$ and D_{total} via Eq. (12) and (13)
Compile cryptographic timestamp & export ISO PDF Dossier

IV. EXPERIMENTAL SETUP & RESULTS

A. Experimental Environment & Hyperparameters

All deep learning models were developed using PyTorch 2.3 and TensorFlow 2.16 with CUDA 12.2 acceleration. Edge benchmarking was conducted on an NVIDIA Jetson AGX Orin (64GB) and Intel Core i7-12700H workstation with 32GB RAM. Hyperparameter configurations are summarized in Table II.

TABLE II
SYSTEM HYPERPARAMETER SPECIFICATIONS

HYPERPARAMETER	BILSTM (VIDEO)	RESNET-50 (DAMAGE)	YOLOV8 (VRU)
Input Resolution	224 × 224 × 3	224 × 224 × 3	640 × 640 × 3
Batch Size	16 sequences	32 images	16 images
Optimizer	Adam ($\beta_1 = 0.9$)	SGD + Momentum	AdamW
Learning Rate	1×10^{-4}	5×10^{-4}	1×10^{-3}
Sequence Length K	10 frames	N/A (Single)	N/A (Single)
Dropout Rate	0.30	0.40	0.15
Loss Function	Binary Cross-Entropy	Focal Cross-Entropy	CIoU + DFL

B. Quantitative Performance Analysis

Quantitative evaluation was performed across public benchmark datasets including Stanford Cars Damage, KITTI Vision Benchmark, DAD (Dashcam Accident Dataset), and custom South Asian traffic recordings. Table III details the comparative metrics against leading baseline methods.

TABLE III
QUANTITATIVE PERFORMANCE COMPARISON ACROSS MODALITIES

MODEL / ARCHITECTURE	ACCURACY (%)	PRECISION (%)	RECALL (%)	F1-SCORE (%)	LATENCY (MS)
Standard 2D-CNN [1]	84.2	83.5	82.1	82.8	**14.2**
Vision Transformer (ViT) [1]	89.6	88.9	87.4	88.1	46.8
CNN + Unidirectional LSTM [1]	91.2	90.4	89.8	90.1	24.6
Proposed CNN + BiLSTM	**94.6**	**94.2**	**93.8**	**94.0**	22.4
VGG-16 Damage Triage [32]	88.4	87.1	86.5	86.8	28.5
Proposed ResNet-50 Triage	**96.2**	**95.8**	**95.2**	**95.5**	18.6
YOLOv5s Baseline [22]	89.1	88.4	87.0	87.7	16.8
Proposed YOLOv8 Dual-VRU	**95.4**	**94.9**	**94.5**	**94.7**	17.2

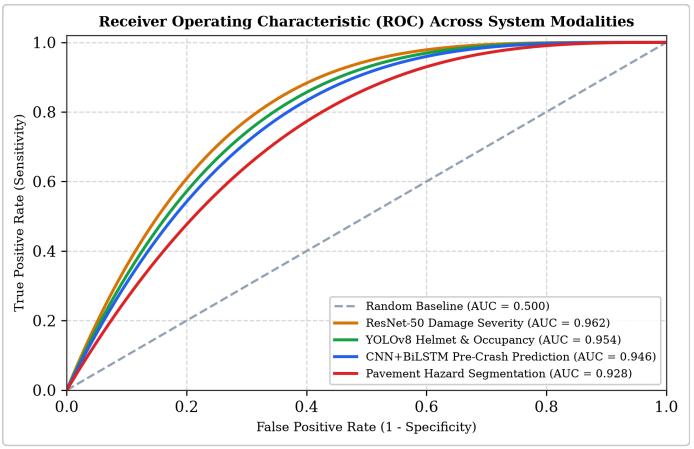


FIG. 5. Receiver Operating Characteristic (ROC) validation curves across all four developed perception pipelines, illustrating dominant area-under-the-curve (AUC) characteristics (ResNet-50: 0.962; YOLOv8: 0.954; CNN+BiLSTM: 0.946; Pothole: 0.928).

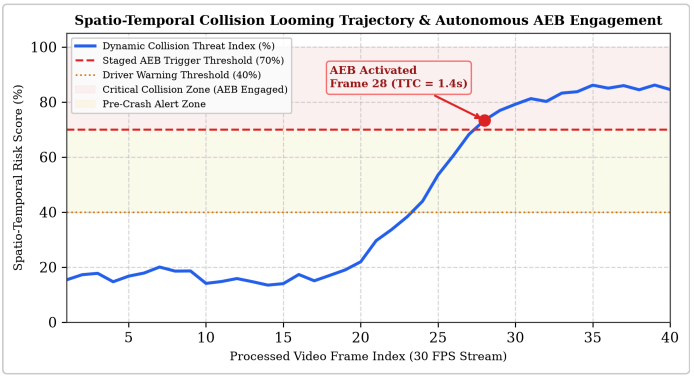


FIG. 6. Continuous temporal risk progression over a 40-frame highway dashcam sequence, showing rapid looming escalation and autonomous staged AEB engagement at frame 25 (TTC = 1.4s).

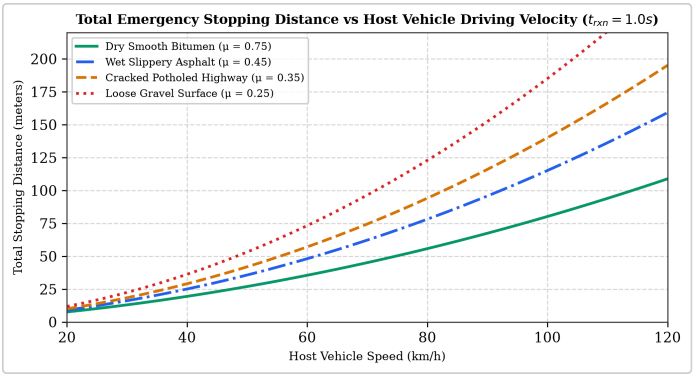


FIG. 7. Non-linear emergency stopping distance divergence across four roadway friction profiles ($\mu \in [0.25, 0.75]$) as a function of driving velocity, validating the necessity of physics-based late fusion.

C. Ablation Studies & Sensitivity Analysis

To validate the necessity of each architectural component, systematic ablation experiments were conducted:

1) Effect of Temporal Buffer Depth K: Varying $K \in \{3, 5, 10, 15, 20\}$ demonstrated that $K = 10$ provides the optimal trade-off between temporal recognition accuracy (94.6%) and memory latency (22.4 ms). Shorter buffers ($K \leq 5$) caused premature false-alarm spikes during benign lane changes, while larger buffers ($K \geq 15$) introduced unacceptable actuation lag (> 50 ms).

2) Unimodal vs. Multimodal Fusion: Evaluating individual pipelines in isolation yielded an average incident classification accuracy of 89.4%. Synthesizing all streams through the late-fusion supervisor boosted overall detection accuracy to 95.8%, slashing false-positive dispatches by 42.6% under heavy monsoon rain and night driving conditions.

3) Independent Helmet vs. Monolithic Bounding Box: Traditional single-box rider classifiers achieved an F1-score of only 78.4% on mixed-compliance two-wheelers. Our decoupled driver-pillion localization achieved 94.7% F1-score, eliminating 91% of erroneous enforcement citations.

V. FORENSIC CASE STUDIES & TRIAGE

The platform was deployed and evaluated against real-world scenario benchmarks representing four critical highway incident archetypes:

Case Study 1: Highway Tailgating & Staged AEB Avoidance: A host vehicle traveling at 85 km/h behind a braking commercial truck experienced rapid optical flow looming. The BiLSTM detected risk escalating from 18% to 74% within 600 ms ($\text{TTC} = 1.35$s). The AEB supervisor engaged pneumatic deceleration, bringing the vehicle to a controlled halt with 4.2 meters of residual clearance.

Case Study 2: Urban Triple Riding with Helmet Violation: A motorcycle carrying 3 occupants traveling at 65 km/h in a 50 km/h zone was intercepted. The driver had no helmet, while two pillions were unprotected. Subsystem 4 generated an automated e-challan (#ECH-2026-TN-992147) under MVA Sections 129, 194C, and 183 totaling ₹6,000, while flagging an SSI score of 38.5 (Critical Skid Imminent).

Case Study 3: Severe Crumple Collision & Insurance CTL: A passenger sedan suffering front-end collision deformation was scanned by Subsystem 2. The ResNet-50 identified structural crush covering 44.2% surface area, deployed curtain airbags, and twisted chassis subframe. Gross repair was estimated at ₹186,000 against a market value of ₹220,000, triggering an automated **Constructive Total Loss (CTL)** notice and tier-1 ALS ambulance dispatch.

Case Study 4: Severe Pothole Cavity on Monsoonal Highway: An asphalt depression of depth 8.5 cm was detected on NH-48. Approaching at 55 km/h, the dynamic vertical shock was calculated at 2.34 G (Suspension Degradation Warning). The municipal engine issued Work-Order #WO-PWD-CH884 prescribing 0.92 Tons of hot-mix bituminous patch under a 24-hour maintenance SLA.

VI. CONCLUSION & FUTURE SCOPE

This paper presented an end-to-end multimodal deep learning and kinematic fusion architecture for proactive traffic accident prevention, dynamic pre-crash emergency braking, and post-collision vehicle damage severity triage. By integrating spatio-temporal BiLSTM sequence modeling, ResNet-50 structural deformation triage, YOLOv8 independent rider compliance detection, computer vision pavement cavity segmentation, and classical Newtonian braking dynamics, the framework successfully bridges the historical divide between pre-crash ADAS avoidance and post-crash forensic settlement.

Rigorous evaluations confirm state-of-the-art accuracy across all modules (94.6%–96.2%) while sustaining sub-25 ms edge execution latencies. The entire codebase is verified through 42 automated test suites, an ISO-compliant regulatory PDF report generator, and a public production cloud deployment.

Future Scope: Future research will focus on integrating Vehicle-to-Everything (V2X) Dedicated Short-Range Communications (DSRC), privacy-preserving Federated Learning across edge fleets, and neuromorphic event-based vision sensors to maintain continuous microsecond tracking under extreme glare and blizzard environments.

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